



Question	Answer	Marks									
1(a)	I 6 unambiguous headings for mass, with correctly displayed units, recorded in Results space, for 1(a) and 1(c) : (Mass of) container + FA 1 / g (Mass of) container (empty or with residual FA 1) / g - Mass of FA 1 (used) / g (Mass of) container + FA 3 / g (Mass of) container (empty or with residual FA 3) / g - Mass of FA 3 (used) / g	1									
	II 6 unambiguous headings for temperature, with correctly displayed units, recorded in Results space, for 1(a) and 1(c) : - Initial / starting temperature / °C - Minimum / maximum / final temperature / °C - Temperature change (or decrease) / °C	1									
	III Precision of readings in (a) and (c) - All four balance readings recorded to same number of decimal places (2 or 3). AND - All four thermometer readings recorded to .0 or .5 °C.	1									
	IV Accuracy mark Calculate the temperature change, ΔT, for Supervisor and each candidate. <table><tr><td>Supervisor temperature change, ΔT/°C</td><td>Award mark IV if δ, the difference between the supervisor ΔT and candidate ΔT, is:</td></tr><tr><td>ΔT ≤ 5.0</td><td>± 0.5 °C</td></tr><tr><td>5.0 < ΔT ≤ 10.0</td><td>± 1.0 °C</td></tr><tr><td>10.0 < ΔT ≤ 15.0</td><td>± 1.5 °C</td></tr><tr><td>15.0 < ΔT</td><td>± 2.0 °C</td></tr></table>	Supervisor temperature change, ΔT/°C	Award mark IV if δ, the difference between the supervisor ΔT and candidate ΔT, is:	ΔT ≤ 5.0	± 0.5 °C	5.0 < ΔT ≤ 10.0	± 1.0 °C	10.0 < ΔT ≤ 15.0	± 1.5 °C	15.0 < ΔT	± 2.0 °C
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Question	Answer	Marks									
1(b)(i)	Energy change = $25 \times 4.18 \times \Delta T$ AND answer correctly rounded to 2–4 sf	1									
1(b)(ii)	$n(\text{carbonate}) = \text{mass used} / 286 \text{ mol}$ AND answer correctly rounded to 2–4 sf	1									
1(b)(iii)	Correct use of data and correct sign for the data collected $\Delta H_1 = \text{(b)(i)} / \text{(b)(ii)} \times 1 / 1000 \text{ kJ mol}^{-1}$ AND answer correctly rounded to 2–4 sf	1									
1(c)(i)	I Correct subtractions in (a) and (c) • Mass of FA 1 and FA 3 used AND • both temperature changes	1									
	II Accuracy mark Calculate the temperature change, ΔT , for Supervisor and each candidate. <table border="1"><tr><td>Supervisor temperature change, $\Delta T / ^\circ\text{C}$</td><td>Award mark IV if δ, the difference between the supervisor and candidate ΔTs, is:</td></tr><tr><td>$\Delta T \leq 5.0$</td><td>$\pm 0.5 ^\circ\text{C}$</td></tr><tr><td>$5.0 < \Delta T \leq 10.0$</td><td>$\pm 1.0 ^\circ\text{C}$</td></tr><tr><td>$10.0 < \Delta T \leq 15.0$</td><td>$\pm 1.5 ^\circ\text{C}$</td></tr><tr><td>$15.0 < \Delta T$</td><td>$\pm 2.0 ^\circ\text{C}$</td></tr></table>	Supervisor temperature change, $\Delta T / ^\circ\text{C}$	Award mark IV if δ , the difference between the supervisor and candidate ΔT s, is:	$\Delta T \leq 5.0$	$\pm 0.5 ^\circ\text{C}$	$5.0 < \Delta T \leq 10.0$	$\pm 1.0 ^\circ\text{C}$	$10.0 < \Delta T \leq 15.0$	$\pm 1.5 ^\circ\text{C}$	$15.0 < \Delta T$	$\pm 2.0 ^\circ\text{C}$
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Question	Answer	Marks
1(c)(ii)	M1: correct expressions: At least 2 correct expressions shown: - energy / J = $30 \times 4.18 \times \Delta T$ - amount / mol = $\frac{\text{mass used}}{106}$ - enthalpy = $\frac{\text{energy}}{\text{amount} (\times 1000)}$ M2: correct use of data negative sign AND correct numerical answer to 2–4 sf	2
1(d)	Correct use of candidate's own data $\Delta H_r = (c)(ii) - (b)(iii)$ AND answer to at least 2 sf	1
1(e)	EITHER Student is not correct AND explanation that acid <u>is still</u> in excess AND temperature rise will be the same OR Student is not correct AND explanation that reaction will be faster OR concentration of $H^+(aq)$ ions is higher AND the quantity of frothing / spitting / acid spray will be greater OR Student is correct plus explanation that reaction will be faster or concentration of $H^+(aq)$ ions is higher AND less (time for) heat loss (to surroundings).	1

Question	Answer	Marks
2(a)	I The following data are recorded: - two burette readings AND titre for the rough titration - initial and final burette readings for two (or more) accurate titrations	1
	II Titre values recorded for accurate titrations, AND correct headings and units in the accurate titration table - initial / start AND (burette) reading / volume - final / end AND (burette) reading / volume - titre OR volume used / added or FA 5 used / added - unit: / cm³ OR (cm³) OR in cm³ (for each heading) OR cm³ unit given for each volume recorded.	1
	III All accurate burette readings are recorded to the nearest 0.05 cm ³ .	1
	IV The final accurate titre recorded is within 0.10 cm ³ of any other accurate titre.	1
	Accuracy (Q) marks Round burette readings to the nearest 0.05 cm ³ . Check and correct titre subtractions where necessary. Examiner selects the best mean titre, using the following hierarchy: - two (or more) accurate identical titres (ignoring any that are labelled 'rough'), <i>then</i> - two (or more) accurate titres within 0.05 cm³, <i>then</i> - two(or more) accurate titres within 0.10 cm³, <i>etc.</i> Calculate the difference (δ) between the candidate's mean titre and the supervisor's mean titre	
	Award accuracy Q marks as follows: V Award if $\delta \leq 0.50 \text{ cm}^3$ VI Award if $\delta \leq 0.30 \text{ cm}^3$ VII Award if $\delta \leq 0.20 \text{ cm}^3$	3
2(b)	Correctly calculates mean titre to 2 d.p. - Candidate must take the average of two (or more) titres that are within a total spread of not more than 0.20 cm³. - Working / explanation must be shown OR ticks must be put next to the two (or more) accurate readings selected. - The mean should be quoted to 2 d.p. and be rounded to nearest 0.01 cm³.	1

Question	Answer	Marks
2(c)(i)	All answers given to parts (c)(ii) , (c)(iii) , (c)(iv) and (c)(v) recorded to 3 or 4 sf.	1
2(c)(ii)	Correct calculation Amount HCl used = $0.110 \times \text{}^{(b)}/_{1000} \text{ mol}$	1
2(c)(iii)	Correct use Amount carbonate = $0.5 \times \text{(c)(ii)} \times 40 \text{ mol dm}^{-3}$	1
2(c)(iv)	Correct use Average $M_r = 5.85 / \text{(c)(iii)}$	1
2(c)(v)	Correct use $106 p + 138.2 (1 - p) = \text{(c)(iv)}$ (where $p = \% \text{ of } \text{Na}_2\text{CO}_3 / 100$)	1

Question	Answer	Marks
FA 7 and FA 8 are $(\text{NH}_4)_2\text{CO}_3$; FA 9 is NaOH		
3(a)(i)	2 • = 1 mark • White solid / powder or white / colourless crystals (at start) • condensation / steam / steamy fumes / water droplets / moisture • (condensation forms on) gentle heating / soon after heating began • No residue (at end) / solid evaporates / solid sublimes / solid reduces in quantity • (Gas / ammonia) turns (red) litmus blue	2
3(a)(ii)	M1: Fizzing / bubbling / effervescence AND EITHER rapid / strong / violent / vigorous (reaction) OR colourless solution formed / colourless solution remains / solid dissolves M2: Gas / CO_2 gives <u>white precipitate</u> with lime water	2

Question	Answer		Marks
3(b)(i)	2 • = 1 mark		7
	<i>test</i>	FA 8	
	Test 1 Ba ²⁺ (aq)	White <u>ppt</u>	
	+ acid	(ppt) soluble / dissolves OR colourless <u>solution formed</u> - fizzing - gas / CO₂ gives white ppt with limewater (credit once only in 3(b)(i))	
	Test 2 AgNO ₃	- White / off-white <u>ppt</u> (ppt) soluble / dissolves/ <u>forms</u> colourless solution	
	+ HNO ₃	- Fizzing - Gas / CO₂ gives white ppt with limewater (credit once only 3(b)(i))	
	Test 3 + FA 9	no change / no ppt / no reaction (in cold) / remains colourless (solution)	
	warm	<u>gas / ammonia</u> turns litmus to blue	
	Test 4 ZnSO ₄	- white ppt (ppt) insoluble in excess	
	Test 5 litmus	red (litmus) goes blue AND blue stays blue / goes darker blue	

Question	Answer	Marks
3(b)(ii)	2 or 3 correct = 1 mark. All 4 correct = 2 marks. FA 8: ions are NH_4^+ and CO_3^{2-} FA 9: ions are 'unknown' and OH^-	2
3(b)(iii)	$(\text{NH}_4)_2\text{CO}_3(\text{s}) \rightarrow 2\text{NH}_3(\text{g}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{g})$	1